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Difference Between Collenchyma and Sclerenchyma

Collenchyma	Sclerenchyma
• Living Cells	• Dead cells
• Contains Protoplasm	• Cells are empty
• Cell walls are cellulosic	• Cell walls are lignified
• Thickening of cell wall is not uniform	• Thickening of cell wall is uniform
• Keeps the plant body soft	• Keeps plant body stiff and hard
• Sometimes it has chloroplast	• Do not have chloroplast

Difference between Fibre and Sclereids

Fibre	Sclereids
• Long cells	• Short cells
• Narrow, Elongated pointed ends	• Usually short and broad
• Occurs in bundles	• Occurs individually or in small groups
• Commonly unbranched	• Maybe branched
• Derived directly from meristematic tissue	• Develops from secondary sclerosis parenchyma cells

Difference between Tracheids and Fibres

Tracheids	Fibres
• Not much elongated	• Very long cells
• Possess oblique end walls	• Possess tapering end walls
• Cell walls are not as thick as Fibres	• Cell wall are thick and lignified
• Possess various types of thickenings	• Possess only pitted thickenings
• Responsible for the conduction and also mechanical support	• Provide only mechanical support

Difference Between Sieve Cells and Sieve Tubes

Sieve cells	Sieve tubes
• Have no companion cells	• Have companion cells
• The sieve areas do not form sieve plates	• The sieve areas are confined to sieve plates
• The sieve areas are not well differentiated	• The sieve areas are well differentiated
• They are elongated cells and are quite long with tapering end walls	• They consist of vertical cells placed one above the other forming long tubes connected at the walls by sieve pores
• The sieve are smaller and numerous	• The sieve pores are longer and fewer
• Found in Pteridophytes and Gymnosperms	• Found in Angiosperms